Theory case · Topic 1 · 2 points · 15 min

C02 — Orbital Fasteners: sorting a day of activity into the functional areas

Orbital Fasteners S.A. makes titanium bolts certified for satellites. On a single Monday: (1) the purchasing team ordered 2 tonnes of titanium bar; (2) the CNC lines produced 40,000 bolts; (3) the sales engineer presented the company at a space-industry fair; (4) the finance department negotiated a bank loan to buy a new thread-rolling machine; (5) it collected an invoice from a satellite integrator and paid the titanium supplier; (6) HR interviewed two quality inspectors; (7) the plant manager set the production targets for the next quarter and checked last month's scrap rate against the plan; (8) the IT team upgraded the traceability database.

Part (a) · 1.25 points

Classify the eight activities into real, financial and management functions

Activity	Function	Group [Slide 13]
1. Ordering titanium	Supply (acquisition of inputs)	Real
2. Producing bolts	Production	Real
3. Trade fair	Marketing	Real
4. Bank loan for a machine	Financing and investing	Financial
5. Collecting and paying	Receiving revenues and paying for inputs	Financial
6. Hiring inspectors	Support (Human Resources)	Real
7. Targets and scrap check	Planning and control	Management
8. Traceability database	Support (Information Systems)	Real

KEY POINTS

- Real: 1 (supply), 2 (production), 3 (marketing), 6 and 8 (support: HR, information systems).
- Financial: 4 (financing and investing), 5 (collecting revenues and paying for inputs).
- Management: 7 (planning and control).

Part (b) · 0.75 points

Explain how the real and the financial cycles relate, and what engineers contribute to the management of a firm like this one

The **financial cycle runs alongside the real cycle** [Slide 13]: the company raises funds (the loan of activity 4), uses them to pay for inputs (the titanium of activity 1, paid in activity 5), turns the inputs into bolts and sells them, collects the revenue (activity 5) and finally returns capital and interest to those who lent the money. Without the financial functions the real cycle stops; without the real cycle there is no revenue to repay the funds.

Engineers bring **project planning, time management and problem solving, attention to detail, experience in data processing, computer skills and technical knowledge** [Slide 15]: exactly what activity 7 (targets and scrap control) and activity 8 (traceability data) require.

KEY POINTS

- The financial cycle mirrors the real one: funds raised → inputs paid → revenues collected → capital and interest returned.
- Engineers: project planning, time management, problem solving, attention to detail, data processing, technical knowledge.